
Bioinformatics

Comparing the Venom Proteins of Phylogenetically Close and Distant Species

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Associate Editor: Prof. Dmitry Korkin

Received on December 12th, 2025

Abstract

Motivation: Snakebite antivenom is currently very specific to various snakes. Envenoming can have incredibly detrimental effects and every second counts when it comes to treatment. At present too much of that time is spent trying to determine the species of snake and locate the correct antivenom. This study aimed to determine if antibodies for one type of venom can be used for other species in order to make more efficient treatments. To achieve this, a list of novel venom proteins was made and docking software was used to check the binding affinity of three synthetic antivenom proteins to the different venom proteins.

Results: The controls showed high affinity binding affinity using the chosen methods, as expected. Therefore, both the methods and controls were effective. The novel proteins generally had higher scores the closer they were to the controls, but the proteins with the highest scores overall was not particularly closely related. More research is needed on this subject.

1 Introduction

In 2023, a total of roughly 2 million cases of snakebite envenoming were reported, with 81,000 – 134,000 cases leading to mortality. Three times that amount caused amputation or permanent disease. Problems related to effective treatment stem from the specificity and availability of many utilized antivenoms. Furthermore, classifying the species of origin using in-depth laboratory-based identification techniques clashes with the rapid time constraints of toxin treatment. Thus far, venom therapeutics are highly specific to individual species, which makes it difficult to treat a variety of patients.

In this paper, we aim to determine if certain antivenoms can be used to treat venom from closely related species. Essentially, venom consists of an assortment of unique proteins, which inhibit vital functions of specific biological functions, such as the nervous system. Current treatment involves using the snakes' blood to harvest antibodies that bind to venom proteins and neutralize them in the infected patient. Simply put, the extent to which antibodies bind to the proteins indicates how effective the antivenom is. To better understand the mechanics underlying antivenom, we will further study the deterministic binding specificity of closely related species' antivenoms to closely related venom proteins.

We hypothesize that there is a direct correlation between the phylogenetic proximity of two species and the binding specificity of an anti-venom that worked on one species to a protein of the other species within the same superfamily.

The main challenge we face in testing our hypothesis is the species requirement of most antivenoms, leading to information about the overall specificity being unknown. Furthermore, information identifying drug targets and binding interactions is not always well classified. Once the mechanics of the protein binding of each antivenom are illustrated, a foundation for comparison is achieved. This will enable the observation of cross-specificity of antivenoms across related species and aim to alleviate response time for patients with snakebite envenomation.

2 Background

Snakebite envenoming is a prominent but under researched tropical disease which has been on the category A neglected tropical disease (NTD) list since 2017. In 2019, the World Health Organization set a goal of halving the mortality rate and disabilities caused by 2030. As of 2019, approximately 63,400 people died from snakebite envenoming, a notable decrease from the estimated over 100,000 deaths in 1998. While there

are many factors that are important to consider in terms of reducing mortality and disability rates, lack of effective treatment is a major contributor. One of the common issues when treating snakebites is the specificity of antivenoms. If a patient does not know which snake the venom came from, it can slow down the treatment process tremendously. For many venoms, the tissue damage progresses quickly, so the efficiency of treatment is essential for preventing death and lifelong disability.

Antivenoms are often developed by infecting a large animal, such as a horse or llama, with snake venom and taking blood samples to extract the antibodies. This process creates extremely specific treatments where one antivenom is developed for every type of snake. This causes many issues for healthcare professionals and patients, both because it requires the patient to see and be able to identify the snake that bit them, and because every hospital has antivenom always stocked for every type of snake in the area. Many tropical countries have a wide range of snake species, all with different venoms, and keeping all antivenoms on hand at all times is not feasible, particularly for developing countries.

At present, there are multiple existing antivenoms that can be used to treat snakebites from multiple species. These are typically made by combining the antibodies from multiple species' venoms into one treatment, such as one developed at the Technical University of Denmark earlier this year. This method is effective and quite promising but does not address the question of whether it is possible to use the same antibodies for multiple venoms. A 2013 study from the University of Costa Rica aimed to determine the similarity of different venom proteins from snakes within the *Bothrops* genus. The study found that there was little correlation between phylogenetic proximity and venom composition, but not that every venom was fully unique. This can be expanded to determine if antibodies from one venom can bind to other venom proteins.

The paper will be organized as follows: In the methods, we outline the procedure we took to create the pipeline of antivenom-venom control complexes, homology proteins from a phylogenetic tree and then comparison of binding capacities from evolutionary related venom proteins to the antivenom control. After the completion of the Methods, details on the results and findings will be expanded upon, with their limitations discussed. Lastly, conclusion will be drawn from the results, along with future directions and project significance.

3 Methods

Anti-Venom and Venom Sourcing

The three anti-venom proteins tested here (from *Torres et al.*) are LNG, B10_CYTX (CYTX), and SHRT binder (SHRT). *Torres et al.* tested these against a *N. kaouthia* long chain α -neurotoxin (LNT) (P01391), *N. pallida* and *N. nigricollis* cytotoxins (P01468 and P0DSN1), and a consensus sequence (SeNTx) of short-chain α -neurotoxins (SNT) from *de la Rosa et al.*, respectively. We used these venom proteins (with the exception of P0DSN1) as a positive control to verify our methods.

A list of possible experimental venom proteins was sourced in three different ways:

- (1) A venom protein used in the *Torres et al.* paper was searched in SCOP, and other proteins within its family were selected.
- (2) Venom proteins used in *Ahmadi et al.* were selected, as this paper dealt with proteins of the same or close families. Those of a different family were searched in SCOP, and other proteins within its family were selected.

- (3) Proteins listed in the Venom Zone database within the broad three-finger toxin family were selected.

A phylogenetic tree was made, composed of the snake species these venom proteins come from (further described in a subsequent section), and proteins were chosen from the list based on their species' proximity (chosen at close, medium, and far distances) to the three positive control species.

Venom and Antivenom Preparation

First off, the PDB identifier of each venom protein structure was imported into UCSF Chimera for necessary processing before docking. This was conducted using the Dock Prep tool under the Tools tab in structure editing. The process involved removing non-essential molecules, such as water molecules, ligands, and ions, that were unrelated to the docking procedure and would only interfere, as these molecules would not be present on the buried surface of the protein interface during experimentation. After each venom protein structure was cleaned, they were saved in the PDB format for HADDOCK compatibility. Following this, in Haddock, the specific amino acid residues participating in the binding pocket were chosen based on the *Torres et al.* Paper and identifier in Chimera using the centroid function on the binding pocket. The antivenom protein structure corresponding to the positive control protein of the same classification was prepared as the docking counterpart. This structure was formatted into the PDB format for consistency throughout docking simulations to assess binding affinity with venom proteins from various species.

Sequence Similarity and Phylogenetic Analysis

The venom protein sequences from various snake species were obtained through various publicly available databases, such as UniProt, SCOP, and Venom Zone. These sequences were aligned by first downloading and combining all FASTA sequences into one file, including the sequences of the positive controls. Minor adjustments were made to the file, specifically to the header of each sequence to better display the names of species involved in the final model. Once the FASTA file is ready, the file was uploaded into MEGA12's multiple sequence alignment tool and the evolutionary relationships were then visualized through a maximum-likelihood phylogram to compare evolutionary distance relative to the controls.

AlphaFold Multimer

Protein-protein complex was first visualized via AlphaFold Multimer. Each anti-venom and venom sequence was inserted into the google colab script and given the appropriate job description. For the specifications, the model type of `alphafold2_multimer_v3` was chosen, along with `rank_num` being increased to 5 and template mode being set to `pdb100`. Every other value was left in the default setting. The resulting zip file was saved to Google Drive and OneDrive. The JSON and PDB files were analyzed in NotePad and ChimeraX, respectively. The PDB files were downloaded for insertion into Rosetta in addition

Docking Simulation with HADDOCK

Protein-protein docking simulations were performed using the HADDOCK 2.4 web server. The docking process occurred in three stages: rigid body docking (it0), it1, and solvent immersion. The Protein and Protein-Ligand files were each uploaded via the saved dock prep

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PDB files from ChimeraX. Then, the active amino acid residues in the binding site were selected from previous literature and the ChimeraX centroid tools. Specifications on the run process were unable to be achieved without the acquisition of an academic license, and as such, the defaults were selected. As finished, the HADDOCK simulation was run producing models and metrics for the anti-venom and venom complex.

4 Results

Three unique de novo antivenom proteins and cytotoxin interactions were utilized as the controls: LNG + P01391, CYTX + P01468 and SHRT + Neurotoxin Complex A. These were selected due to their proven efficacy in previous studies. An AlphaFold Multimer prediction was further run to test the efficacy of the specific protein-protein interaction and create a baseline model for visualization efforts in Chimera.

Table 1. AlphaFold Multimer Results of Control Venom and Anti-venom Interactions. Three distinct De Novo antivenoms were bound with their primary venom target and predicted through AlphaFold Multimer. Sequences acquired from PDB publications. Baseline pLDDT, ptm and iptm were produced for each interaction.

	pLDDT	ptm	iptm
LNG + P01391	88	0.83	0.85
CYTX + P01468	83.2	0.771	0.794
SHRT + ScNTx Consensus Sequence	86.2	0.806	0.813

Table 2. HADDOCK Metrics of Control Protein Complexes. Three distinct De Novo Venom Binder AlphaFold multimer binding predictions, with their control venom, were analyzed through HADDOCK. Quality and Realism of each protein-protein interaction were quantified using HADDOCK score, Cluster Size, RMSD, Van der Waals Energy, Restraints Violation Energy, Electrostatic Energy and Z-score.

	HADDOCK Score	Cluster Size	RMSD	Van der Waals Energy	Restraints Violating Energy	Electrostatic Energy	Z-Score
LNG + P01391	-66.5 +/- 5.3	10	12.6 +/- 0.3	-27.2 +/- 3.3	23.8 +/- 14.0	-83.0 +/- 33.0	-1.6
CYTX + P01468	-73.7 +/- 5.8	19	9.8 +/- 0.0	-37.6 +/- 3.6	223.3 +/- 43.1	-387.4 +/- 33.8	-1.1
SHRT + ScNTx Consensus Sequence	-76.7 +/- 4.6	19	8.8 +/- 0.2	-42.3 +/- 2.2	256.4 +/- 34.5	-401.2 +/- 33.4	-1.3

To confirm that the respective protein-protein interaction would occur in addition, we observed their HADDOCK Metrics. Accumulation of their Interface Score, RMSD, and TM-score enabled a way to ensure the binding energies were realistic with their quality sufficient. This complimented the AlphaFold predictions and granted a solid baseline for comparison with the homologous venoms.

Each HADDOCK simulation speculated the binding atoms unique to their venom. The LNG + P01391 Control matched the amino acid residues Cys41 in ScNTx and the Tyr45 and hydrogen backbone binding. The LNG + P01391 Control matched the Glu69, Tyr40 and Tyr49 of the LNG structure to that of the Arg33 in loop II of the cytotoxin amino acid residues. Lastly, the SHRT + Neurotoxin Complex utilized the interaction network between loop II and III.

The HADDOCK scores indicate a successful predicted binding for all three of them. This shows that the complex between the venom and antivenom is energetically favorable in nature. The Z-scores give further confidence in that regard. The Van der Waals energy and restraints violation energy display that the bonds formed are stable as well.

Table 3: Rosetta Scores of Antivenom-Venom Controls; LNG and P01391, CYTX and P014688 and SHRT and Neurotoxin Complex A Display of their respective Total Score, dsfl_fa13, fa_atr and fa_elec are shown below.

	Score	dsfl_fa13	fa_atr	fa_elec
LNG + P01391	1289.402	-6.605	-785.014	-785.014
CYTX + P01468	-1.938	-6.305	-853.64	-220.794
SHRT + ScNTx	174.176	0	-511.274	-125.201

Furthermore, their Rosetta scores were analyzed based on quantitative comparisons in the future. Each Antivenom-venom pair was subjected to such analysis. The scores indicated CYTX + P01468 were the most stable and effective out of the three controls. However, all three performed substantially well in their Total Score, dsfl_fa13, fa_atr and fa_elec. The combination of the AlphaFold Multimer, HADDOCK and Rosetta indicate the complex formed in both realistic and stable and can be used for future comparisons to the homologous proteins.

4.1 Phylogenetic Tree

Once the controls were isolated, we determined the homologous venom proteins.

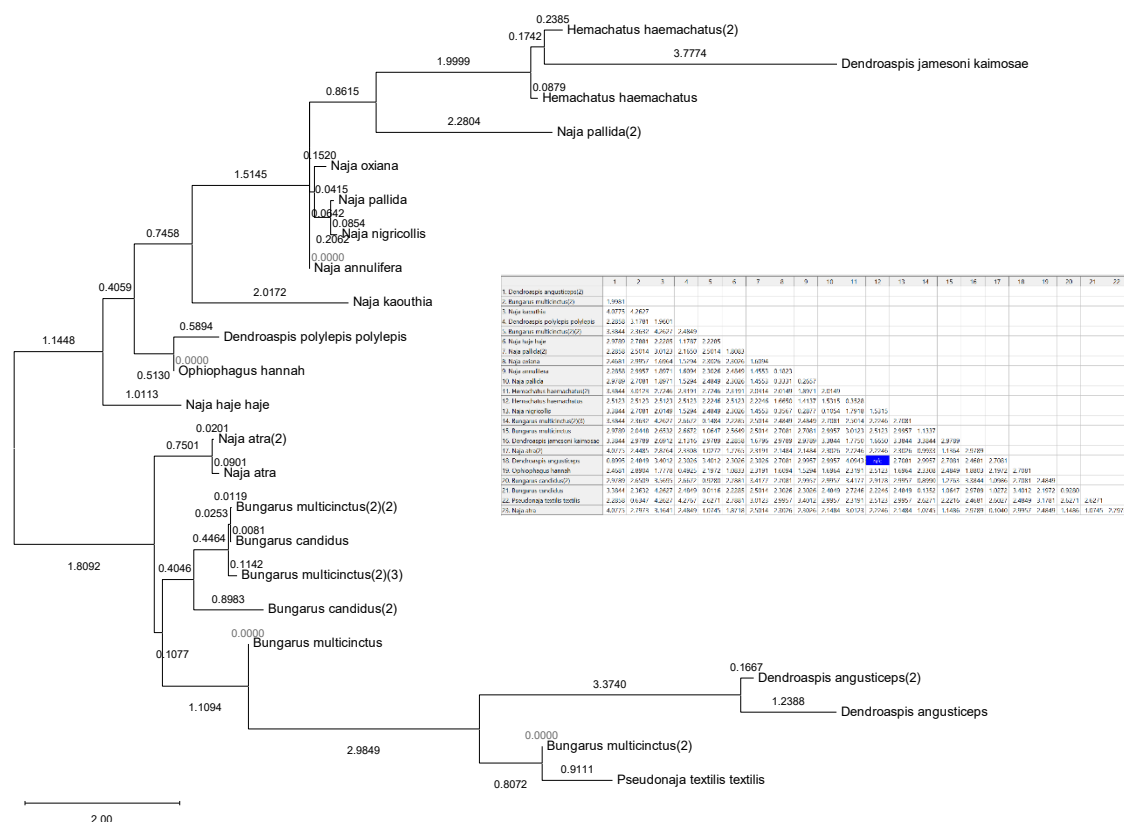


Figure 1. Pairwise distance chart between the various species' venom protein sequences. Phylogram of venom proteins with evolutionary distances displayed. The pairwise distance chart was utilized to create a phylogenetic tree of the various species proteins evolutionary distances to one another.

4.2 Related Species Venom

Table 3. AlphaFold Multimer Results of Novel Venom and Control Antivenom Binders. Each distinct interaction is classified by their respective pLDDT, ptm and iptm scores.

	pLDDT	ptm	iptm
P60301 + CYTX	80.8	0.701	0.861
P01451 + CYTX	79.8	0.714	0.623
P01471 + CYTX	90	0.824	0.809
P81658 + CYTX	66.6	0.508	0.179

Each species was subject to AlphaFold Multimer modeling predictions. P01451 was the most closely related species. P81658 was the least closely related species, and the other two species fell in between.

Table 4. Rosetta Scores of AlphaFold Multimer Results of Novel Venom and Control Antivenom Binders. Each distinct interactions Score, dsIf_fa13, fa_atr and fa_elec scores are displayed respectively.

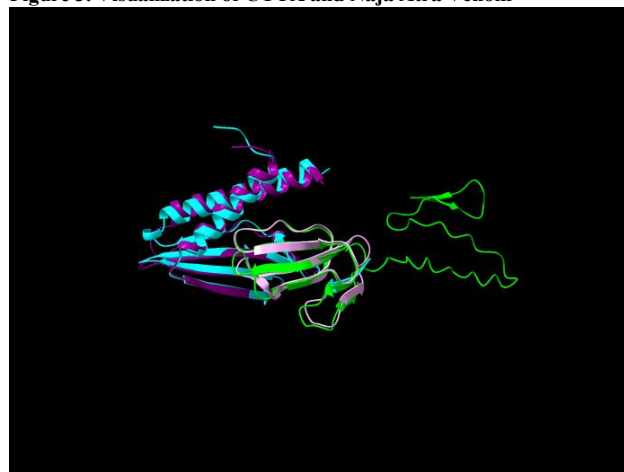
	Score	dsIf_fa13	fa_atr	fa_elec
P60301 + CYTX	7464.434	17.069	-1060.545	-268.731
P01451 + CYTX	637.031	12.859	-892.997	-244.7
P01471+	431.663	15.022	-876.348	-236.752

CYTX				
P81658	+	2670.806	11.836	-939.45
CYTX				-248.032

4.3 Visualization of the Proteins

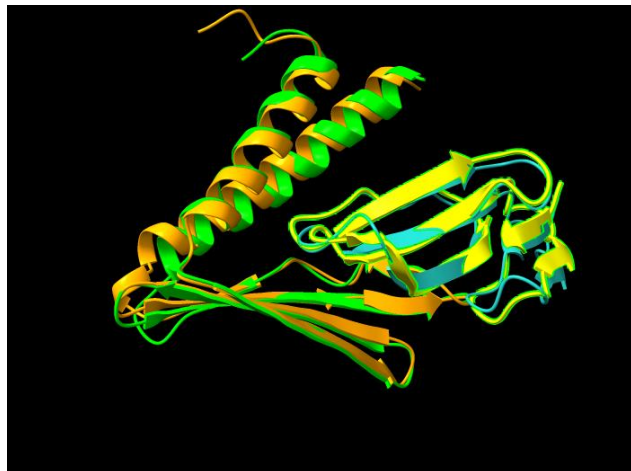
The PDB files from the AlphaFold results were then uploaded to Chimera and superimposed over the control protein complexes using the "Matchmaker" tool.

Figure 3. Visualization of CYTX and Naja Atra Venom



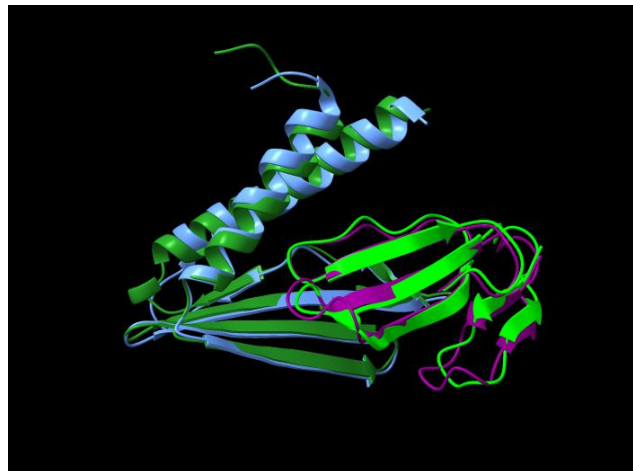
The novel protein complex was aligned with the CYTX control complex. The blue and green proteins make up the novel protein complex, and the pink and purple proteins make up the control. The blue and purple proteins are CYTX, and the pink and green proteins are the venom proteins.

Figure 4. Visualization of CYTX and Naja Oxiana Venom



The same methods were used here as above. In this image the green and yellow proteins are the control, and the blue and orange proteins are the novel proteins, with orange and green showing CYTX and blue and yellow showing the venom proteins. The alignment was much closer for this protein complex than for the last one. This was expected because Naja Oxiana is the closest species to control species phylogenetically. The protein involved is almost exactly the same.

Figure 5. Visualization of CYTX and Hemachatus haemachatus



This visualization used the same methods as the previous two. The dark green and purple show the novel protein, and the blue and light green show the control complex. CYTX is shown in blue and dark green, and the venom proteins are light green and purple. This species was listed further than the previous two on the phylogenetic tree but the complexes actually align even more closely.

Figure 6. Visualization of CYTX and Dendroaspis angusticeps

This visualization also used the same methods. The control complex is shown in orange and red and the novel protein complex is shown in blue and green. CYTX is blue and orange here and the venom proteins are red and green. This species was the furthest from the control as expected the alignment was not as close as the other species were.

5 Conclusion

The results for the controls were as expected. All methods showed high binding affinity for the antivenom and venom proteins. This indicates that the chosen methods are effective. This removes a high percentage of possible errors in future work on this topic. The results for the AlphaFold multimer alignment for the chosen novel proteins showed the close to the expected results as well, with more similar species; species that are closer on the phylogenetic tree, having higher scores for binding to the antivenom proteins. However, one of the species that was less related showed resulted in higher scores than any of the other novel proteins and even higher than many of the control scores. These results were further supported by the visualization and the Rosetta results. More research is needed to determine the cause, and more species are needed to obtain significant results. The current results are insufficient to support the hypothesis.

Many of the methods chosen were both effective and efficient, but HADDOCK took a significant portion of time, hence only being used for the controls at this stage. For future research, Rosetta will be used to analyze the proteins and the binding more thoroughly since Rosetta provides more relevant data and is faster.

For continuing this work, we would use these same methods on a wider range of proteins and species. We would look at more antivenom proteins than just the three synthetic ones as well. The scale of this project was rather small, which prevented a more thorough investigation of our hypothesis. We would also use Schrodinger as a more specific method for checking the docking. We had initially intended to use it for this project but were unable to due to the cost of the software.

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